STEVEN LIM

R&D; Manager | Industrial Software & Systems Integration | Engineering Leadership

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PROFILE

Engineering leader and hands-on systems integrator with 8+ years of experience bridging C#/.NET and Python software, MES/SFIS, SAP interfaces, robotics, industrial controls and production operations. Leads multidisciplinary R&D; engineering teams while contributing to architecture, machine connectivity, commissioning and client delivery. Experienced with Japanese and Chinese customers and complex manufacturing environments.

PROVEN IMPACT

- Lead a multidisciplinary engineering organization of 40+ people, with 12 direct reports.
- Led automation improvements that reduced direct operating manpower on a production line from 16 to 4 while supporting production objectives.
- Contributed to international project delivery, repeat customer engagements, and department credibility with executives.
- Introduced a two-shift operating arrangement to increase production capacity and support improved business performance.

EXPERIENCE

R&D; Manager / Manager of Process Information 05/2025 – Present

PT Sat Nusapersada Tbk · Batam, Indonesia

- Lead software, automation, mechanical, electrical and fabrication teams from concept through design, integration, UAT, commissioning and production handover.
- Coordinate client technical and commercial discussions, including during difficult financial periods, to support project continuity.
- Deliver connected manufacturing solutions linking equipment, robots, sensors, MES/SFIS and production databases.
- Support cross-cultural customer relationships, technical delivery and repeat-project opportunities.

System Analyst & Developer 12/2024 – 05/2025

PT Sat Nusapersada Tbk · Batam, Indonesia

- Designed reusable C#/Python/PostgreSQL application components for industrial automation and manufacturing integration.
- Developed production integration workflows linking sensors, embedded devices, databases and automated inspections using YOLO/OpenCV.

Senior System Developer 01/2022 – 12/2024

PT Sat Nusapersada Tbk · Batam, Indonesia

- Developed machine-monitoring/OEE dashboards and alerting for visibility into downtime and equipment state.
- Modernized MES-related systems with C# and SQL Server, including serial-COM equipment interfaces and middleware to customer test systems.

Junior System Developer 07/2018 – 01/2022

PT Sat Nusapersada Tbk · Batam, Indonesia

- Developed factory ESD gate-check and monitoring tools using C#, sensors, Arduino and PostgreSQL.
- Implemented environmental monitoring for 10+ production zones using Python, PHP and PostgreSQL.

SELECTED ENGINEERING SYSTEMS & PROJECTS

SAP–MES Integration

Integrated SAP and MES via C#, SQL, and .NET APIs; delivered connectivity between enterprise and manufacturing systems.

Customer Machine–MES Integration

Connected external/customer machines to MES using serial-COM readers and C# processing for shop-floor data exchange.

Production-Line Automation & Robotics

Led multidisciplinary integration of SCARA/cobots, PLCs, sensors, MES and production equipment; one project reduced direct staffing from 16 to 4.

Hikvision / YOLO Smart Parking Prototype

Developed a basic camera-based parking vacancy detection flow with Hikvision CCTV, Python and YOLO.

Factory IoT and Health Screening

Integrated LM35 sensors with Raspberry Pi, Python, PostgreSQL and PHP. Built COVID-era screening/entry-validation components using Python, Raspberry Pi and GraphQL.

Hikvision Helper Migration and Messaging Workflows

Ported Hikvision helper logic from C# to Python; developed JavaScript meeting-minute delivery via Telegram.

Factory Black Box — Independent Project, In Progress

Developing a machine event recording and replay / root-cause analysis platform using ASP.NET Core/.NET 10 and React/TypeScript. PostgreSQL and MQTT/OPC UA integration planned.

Spare Parts and Inventory Controller — Independent Project

Self-directed spare-parts and inventory control application concept/project.

TECHNICAL EXPERTISE

Software: C#, .NET, ASP.NET Core Web API, Python, JavaScript, TypeScript, PHP, REST APIs, GraphQL, Windows desktop apps

Data & development: SQL Server, PostgreSQL, Entity Framework, Git, GitHub, Bitbucket, Visual Studio, VS Code, Postman, Docker

Industrial & embedded: MES/SFIS, SAP interfaces, serial communication, Modbus RTU, Panasonic FP0/FP2 PLCs, Arduino, Raspberry Pi, sensors

Robotics & vision: SCARA, cobots, machine integration, YOLO, OpenCV, Hikvision CCTV, Keyence vision exposure

Leadership: R&D;/team leadership, contract discussions, client delivery, commissioning, UAT, cross-functional coordination, operational improvements

EDUCATION & LANGUAGES

Bachelor of Electrical Engineering — Universitas Internasional Batam, Indonesia (2018)

Languages: Indonesian (native); English (TOEIC 880 / professional working proficiency).

Mobility: Open to relocation to Germany and other European countries; appropriate work authorization required.